

TIN DANG HUYNH TRUNG

Senior Software Engineer – Backend, AI Platforms & Distributed Systems

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Senior software engineer with eight years of building and running production backend systems in Python, Rust and TypeScript across fintech, conversational AI, computer vision and data platforms. Sole architect on several greenfield platforms, technical lead on teams of up to fourteen. I lean toward systems where correctness is enforced by structure rather than convention: append-only ledgers, multi-tenancy that fails closed, money that cannot move twice.

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Technical Lead

Trustify Technology

Jan 2024 – Present

Ho Chi Minh City, Vietnam

Fintech, real-estate and intelligent-transport platforms for Yoma Group, Yoma Bank, Wave Money and Trustonic across Vietnam and Myanmar. I architect and lead client rebuilds, and wrote the company's engineering career ladder.

- **Estate-finance platform** • *Yoma Group – sole engineer & architect*

Rebuilt a live property-management and treasury system off PHP/Laravel onto one append-only double-entry ledger, so every balance can be re-derived and audited, with per-customer cutover, an outbox and standing reconciliation. Graded 235 legacy business rules against it and published the gaps before anyone moved.

Python 3.12 • FastAPI • async SQLAlchemy 2 • PostgreSQL (schema-per-tenant) • Redis • Next.js • AWS ECS • Envoy

- **Card payments & PCI boundary** • *Yoma Group, Wave Money*

Cardholder data in its own service, database and network, with a CI-guarded import ban keeping the rest of the platform out of PCI scope. Integrated 2C2P with idempotent callbacks and three-way reconciliation.

Python • FastAPI • PostgreSQL • 2C2P • Docker

- **Urban traffic intelligence** • *Ho Chi Minh City – architect & sole engineer*

Three-tier platform for 400+ cameras where raw video never leaves the edge: Jetson nodes cut backhaul bandwidth 75%, YOLO models track vehicles at a P95 of 85 ms, events reach the data centre over mutually authenticated Kafka. Load-tested to 71K events/s before adopting it; plates pseudonymised for GDPR.

Python • FastAPI • Kafka • PyFlink • TensorRT • Triton • K3s • ArgoCD • Jetson • DuckDB

- **Microfinance lending & resident services** • *Yoma Bank, Star City*

Built Yoma Bank's lending MVP and led its bilingual staff-culture survey. On the Star City resident platform I owned the money, authentication and payment paths for a nine-person team, closing auth-enumeration, TOCTOU and privilege-escalation findings before launch.

Java 21 • Spring Boot 3.3 • jOOQ • Kafka • Flyway • Python • FastAPI • TypeScript

- **Device-management migration** • *Trustonic*

Made the cutover reversible by owning the per-customer rollback across PostgreSQL and DocumentDB, and tuned the migration jobs: vectorised batch validation, broadcast joins, keyed DynamoDB reads so a run never scans a live table, measured Redis partitioning.

PySpark • AWS Glue • Airflow • DynamoDB • PostgreSQL • DocumentDB • Redis • boto3

Founder & Principal Engineer

Pilot Space (open-source AI infrastructure)

Jan 2026 – Present

Remote

- **Moon** • *Redis-compatible in-memory store, written from scratch*

RESP2/RESP3 and 250+ commands so existing clients connect unchanged, with native vector, BM25 and graph search under MVCC. Released 30 times to v0.8.9; measures 1.62× GET and 1.32× SET against Redis 7.0 with 8–22% less memory per key, published with the cases it loses.

Rust • Tokio • io_uring • crates.io • PyPI

- **Lunaris & Helios** • *agent memory and agent harness*

Lunaris recalls across 100K documents per scope in 19–22 ms at the median, guarded by a CI gate that fails any 5% regression. Helios is an agent harness with 12 provider backends, a native MCP client and server, WASM-sandboxed extensions, and circuit-breaker policy on every outbound call.

Rust • Tokio • MCP • Wasmtime • npm

- **Hydroa** • *multi-tenant LLM gateway*

Envoy in front of FastAPI, six providers behind per-tenant and per-provider circuit breakers, customer keys encrypted at rest, and a cost ledger reconciled against what providers actually billed.

Python • FastAPI • Envoy • PostgreSQL • Kubernetes • OpenTelemetry

- **ADD** • *published development method*

A spec-first, red/green method for building software with AI agents, published as @pilotspace/add (npm) and pilotspace-add (PyPI) at about 1.2K downloads a month; the working method on the platforms above.

CTO – Co-founder

ClickAI Viet Nam

📅 Jun 2024 – Oct 2025 (concurrent)

📍 Ho Chi Minh City, Vietnam

Co-founded ClickAI (clickai.vn) and built its conversational AI platform from scratch: multi-modal chat, image and audio generation and community, grown to 40K+ users across Southeast Asia within four months. As CTO I set the architecture and engineering standards, led a product team of twelve to fourteen, and owned billing, identity and deployment.

- **Billing & metering** • *sole author*

Wallets, plans, promotions and referral commissions over a month-partitioned ledger metering every LLM call. A reconciliation check I wrote found provider billing had drifted more than 30% from what we metered; I fixed the ledger model and indexes, keyed idempotency on the source event so retries cannot double-count, and made the check a standing job.

Python · FastAPI · PostgreSQL (partitioned) · Redis

- **Model platform** • *20+ providers*

An LLM orchestration layer with a custom provider alongside 20 third-party ones, routing with failover, vector memory and reasoning-model support; policy-based routing cut API spend by 20%. Generation fans out to ComfyUI, Replicate, fal.ai, ElevenLabs and OpenAI behind circuit breakers.

Python · FastAPI · Weaviate · Qdrant · FalkorDB

- **Production, identity & web** • *99.9% uptime, 1K+ concurrent sessions*

Authored the 13-service deployment with cert-manager and autoscaling, holding sub-2s responses. After a saturation incident I moved PostgreSQL onto a three-node Patroni cluster, rolled back a bad release my instrumentation caught before any customer did, and cut hosting cost 50%. Also shipped OAuth, TOTP and refresh-token rotation, and the web client's community and creative features with SSE streaming, workspaces, RBAC and bilingual localisation.

Kubernetes · Helm · Azure · Patroni · Prometheus · Next.js 14 · TypeScript

Earlier Experience (2018 – 2024)

- **Senior Data Engineer**, Asia Commercial Bank (ACB) • Aug 2023 – Jun 2024

Enterprise data lake at 5M+ daily transactions. A data-quality framework I built across six dimensions cut manual verification by 80%; my APIs served 100+ internal applications.

Cloudera · PySpark · Dagster · Impala · Iceberg · FastAPI

- **Software Data Engineer**, Vietmoney Corp • Mar 2021 – Aug 2023

Fintech appraisal and lending at 10M+ daily transactions, 99.9% uptime. Price forecasting at 96% accuracy cut appraisal from 15 minutes to 5 seconds; Vietnamese NER reached 96% F1; a distributed crawler doubled throughput across 50+ sources.

Python · PyTorch · BERT · Scrapy · Pub/Sub · BigQuery · Airflow · Kubernetes · GCP

- **Software Developer**, Dai Phat Solutions • Aug 2019 – Mar 2021

CCTV analytics at 15 FPS for 20+ clients cut reported incidents by 40%; face recognition at 92% accuracy over 1,000+ employees removed 90% of a manual process.

Python · PyTorch · OpenCV · FAISS · Kafka · Redis · FFmpeg

- **Full-stack Developer**, LogiGear Corporation • Jun 2018 – Mar 2019

Web3 wallet, smart-contract and liquidity-pool management on Ethereum with MetaMask, in Node.js and React.

OPEN SOURCE

- **Moon**, Lunarix and **ADD** at github.com/pilotspace (described above).
- **yowo** (PyPI) — production YOLO inference and export across TensorRT, ONNX Runtime, CoreML and OpenVINO, with ByteTrack and CLIP-ReID tracking. **fastapi-multi-tenant**; **rjson**, Rust-backed JSON for Python; **face-anti-spoofing**, founded at Vietmoney.

TECHNICAL SKILLS

Languages & backend: Python (typed, async), Rust, TypeScript / Node.js, SQL, Java; FastAPI, SQLAlchemy 2, Next.js / React, clean & hexagonal architecture, CQRS, event sourcing, multi-tenancy, outbox & idempotency, PCI boundaries, OAuth / OIDC

AI, data & cloud: PyTorch, YOLO, TensorRT, Triton, LLM gateways, RAG, MCP; Kafka, PyFlink, PySpark, AWS Glue, Airflow, BigQuery; PostgreSQL (Patroni HA, RLS, partitioning), Redis, MongoDB, DynamoDB, Weaviate, Qdrant; Kubernetes, Helm, Azure, AWS, GCP

EDUCATION, TEACHING & CERTIFICATIONS

BSc Computer Science (High-Quality Program), Ton Duc Thang University, 2015–2019, GPA 3.2. **Teaching:** Instructor, Design Patterns, Ton Duc Thang University (2022); Mentor, MindX (2023); Speaker, AI Workshop, FPT University (2025). **Certifications:** Google Cloud AI Agents (2025); AWS Machine Learning Foundations, Udacity (2022); Scrum, Axon Active (2017). **Languages:** English, Vietnamese.